

Ibad Ullah Zuberi

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Education

Brunel University London

London, UK

BEng (Hons) Computer Systems Engineering, with Industrial Placement

Sep 2023 – 2027 (expected)

- **Year 1 (2023–24): A+ (First Class)** **Year 2 (2024–25): A+ (First Class)**
- Topper (1st of 30), Course Representative, Member of IET and DSS Brunel

Experience

BMW Group

Hams Hall, UK

Logistics Planning Intern (Controls & Automation)

Jul 2025 – Sep 2026

- Programmed **PLC ladder logic (Siemens TIA Portal V17, WAGO e!COCKPIT)** for automated material-flow systems, including status lamps that give staff live visibility of autonomous transport robot operation.
- Projected a **15%** throughput gain on a semi-automated forklift system by configuring devices, diagnosing serial-communication faults one variable at a time, and writing deployment and recovery procedures.
- Supported **Industry 4.0** projects by authoring technical specifications and functional requirements issued to suppliers for quotations.
- Doubled KPI coverage from 13 to **27** with an Oracle APEX logistics dashboard fed automatically from several source systems, presented to **60+** colleagues.
- **Technologies:** Siemens TIA Portal V17, WAGO e!COCKPIT, Oracle APEX, SQL, SAP, Excel/VBA

EY (Ernst & Young)

London, UK

Data Analyst

Jul 2024 – Sep 2024

- Enhanced decision accuracy by analysing **20,000+** data entries in SAP Analytics Cloud.
- Reduced planning time with SAC predictive modelling for data-driven forecasts, and cut data interpretation time by creating **5 interactive dashboards**.
- Optimised key business metrics by designing and implementing custom calculations.
- **Technologies:** SAP Analytics Cloud (SAC), Microsoft Excel, Microsoft Office

Cloud Nebula Enterprises

Web Developer

Jun 2024 – Sep 2024

- Boosted financial decision-making efficiency by developing a web-based data analysis tool that handles **60,000+** entries.
- Reduced data interpretation time by **25%** by implementing interactive visualisations.
- Translated business requirements into technical solutions by working closely with 3+ domain experts.
- **Technologies:** Python, Pandas, Dash, NumPy, HTML/CSS, Machine Learning Basics, Finance Basics (Stocks, Options, Commodity)

Projects

Final-Year Project | AI route optimisation, BMW Hams Hall

Sep 2026 – Apr 2027

- Developing an AI model to sequence plant delivery (taxi) routes dynamically, aiming to cut loading time and raise pallets per trip.

Pegboard | Next.js, PWA | [Live site](#) | [GitHub](#)

Live

- Launched a family chore and pocket-money app where parents post priced jobs, children claim them with photo evidence, and parents sign off from one queue; installable as a PWA with a PIN kiosk mode for shared tablets.

Hack the Wallet | Next.js, Starknet, Gemini 1.5 Flash | [GitHub](#)

Apr 2025

- Won **1st place** at the Encode AI London Hackathon (main track plus two sponsor bounties) with a Starknet game where players persuade an AI to return locked crypto, using Gemini 1.5 Flash trust scoring, voice input/output and on-chain rewards on Sepolia.
- Connected Argent X and Braavos wallets to smart contracts that mint a 100-token airdrop and handle betting and payouts, in a retro pixel-art interface.

Go Fish | Next.js, Solidity, Polkadot | [GitHub](#)

Apr 2025

- Reached the finals of the EasyA x Polkadot Hackathon London with a payment platform for fishing import/export where exporters receive **97.5%** of trade value instantly, using smart contracts and stablecoin integration.

- PIC16F18877 Digital & Analogue Interfacing** | XC8 C, MPLAB X, E-blocks2 Jan – Mar 2025
- Completed 8 hardware-tested exercises, including a 4-digit multiplexed 7-segment counter (0000–9999), an HD44780 LCD in 4-bit mode, and 10-bit ADC readings from the on-board light sensor shown as a voltage.
 - Built a five-switch priority controller (bouncing-ball and sweep LED patterns) documented with a flowchart, using register-level PORT/LAT/TRIS control.
- StreamFlow** | TypeScript, React, Rust, Radix | GitHub 2024
- Won **1st place** at Radix Hack with a decentralised creator-tipping platform on the Radix blockchain that sends viewer tips directly to streamers.
- LuffaBot AI Assistant** | Python, LangGraph, FastAPI | GitHub 2024
- Won **1st place** at Brunel University Hack in under 24 hours with a LangGraph multi-tool agent on the Luffa messaging platform that runs group votes and generates images from chat.
- Foodo-Baggins** | Next.js, TypeScript, MongoDB, Ollama | GitHub 2024
- Placed **3rd** in the Verdn environmental track at Royal Hackaway v8 with an AI nutrition tracker that analyses meal photos and gives calorie estimates and personalised advice.
- PIC Assembly Programming** | PIC16F18877 assembly, MPLAB X Oct – Dec 2024
- Completed 6 hardware-tested assembly exercises, including cycle-counted delay routines at 20 MHz tuned to a 1 Hz flash, a three-switch priority LED sequencer and lookup-table 7-segment drivers.
 - Wrote LCD initialisation and message subroutines and a multi-digit display driver, working directly with banked registers and computed jumps.
- Digital Systems Design** | Verilog HDL, Quartus, MAX7000 CPLD Oct – Dec 2024
- Designed a programmable pulse generator (50 μ s pulses, repeat period selectable from 100 to 800 μ s) using a top-down FSM method, then re-implemented it in Verilog at gate and functional level.
 - Built a three-module system (digit generator, weighted-BCD encoder, parity generator) integrated through a gate-level top module and verified in waveform simulation.
- Weather Monitoring System** | Raspberry Pi Pico, BME280, Python, REST API Oct 2024
- Achieved **95%** sensor accuracy in a real-time monitoring system built on a Raspberry Pi Pico with a BME280 sensor.
 - Served readings through REST APIs that support concurrent requests.
- Automated Object Retriever** | Arduino, Ultrasonic Sensor, Embedded C Jan – Mar 2024
- Reached **95%** item-retrieval accuracy and cut manual retrieval time by **70%** by leading development and testing of an autonomous robot.
 - Built hardware and software together from scratch, programming an Arduino with ultrasonic sensing in embedded C.
- Voice Assistant** | Python, Pyttsx3, SpeechRecognition, Wikipedia API Feb 2024
- Reached **92%** recognition accuracy across 6 voice commands in a Python voice assistant built at Royal Hackaway V7.
- Electronics Laboratory: RC Transient Analysis** | Oscilloscope, function generator, multimeter Dec 2023
- Verified RC charge/discharge theory ($\tau = 100$ s) by logging capacitor voltage every 30 s over 5τ , explaining deviations from component tolerance, wire and switch resistance, and measurement limits.
 - Used a bench power supply, function generator, multimeter and oscilloscope to measure I/V characteristics, RMS voltage and signal attenuation.

Achievements

Hackathons: 1st Encode AI London Hackathon (Apr 2025) | 1st Brunel University Hack (2024) | 1st Radix Hack (2024) | 3rd Royal Hackaway v8, Verdn track (2024) | Finalist, EasyA x Polkadot Hackathon London (Apr 2025)

Technical Skills

Controls & Automation: PLC Programming (Ladder Logic), Siemens TIA Portal V17, WAGO e!COCKPIT, LabVIEW, Industry 4.0

Embedded & Hardware: Embedded Systems, Hardware-Software Integration, PIC16F18877 (assembly, XC8 C), Arduino, Raspberry Pi (Pico), FPGA Programming, Sensor Integration, Soldering, Breadboarding

Programming: C, C++, Python, Assembly, Verilog HDL, MATLAB, Java, TypeScript, SQL, VBA

Design & Tools: CAD (Fusion 360), Quartus, PSpice/OrCAD, MPLAB X, Git, Oracle APEX, SAP, Pandas, NumPy